

Unmanned Aerial Vehicle with Human Detection and People Counter Using YOLO v5 and Thermal Camera for Search Operations

Jewel Kate D. Lagman
School of Electrical, Electronics and
Computer Engineering
Mapúa University
Manila, Philippines
jkdlagman@mymail.mapua.edu.ph

Alden B. Evangelista
School of Electrical, Electronics and
Computer Engineering
Mapúa University
Manila, Philippines
abevangelista@mymail.mapua.edu.ph

Charmaine C. Paglinawan
School of Electrical, Electronics and
Computer Engineering
Mapúa University
Manila, Philippines
ccpaglinawan@mymail.mapua.edu.ph

Abstract— When a disaster or accident occurs, the first thing that must be done is search operations; this must be done carefully and discretely using a well-defined procedure. Its purpose is to help people who are in immediate danger. However, conducting search operations by a person, especially in a risky location, is not safe due to limited equipment and dangerous scenarios. This study developed an Unmanned Aerial Vehicles (UAV) prototype that captures even with an overview head image for the search operation and detects humans in the image. After detecting the presence of a human in an image, the prototype counts the actual people in the frame and displays the total number of people discovered in real time. Suppose the same person was detected twice but in a different frame, it will be still counted on the overall person counts. In object detection, You Only Look Once Version 5 (YOLOv5) is used as an algorithm. Thermal camera will be added for more accurate results from the detected person.

Keywords— Search, YOLOv5, Tello Drone, Raspberry Pi, FLIR Lepton

I. INTRODUCTION

Search and rescue operations are carried out in close collaboration with the community and as a team effort. When a community is affected by natural disasters the rescue operation takes place to save people that may be trapped by falling debris. Unmanned Aerial Vehicles (UAVs) can help with search operations due to being fast and capable of autonomous behavior, allowing them to carry out actions that would be difficult for humans to carry out at low cost and can protect rescuers' lives out of extreme danger. It can also explore the area and collect evidence about the whereabouts of a missing person.

A detailed examination of a large amount of recorded material was done using a DJI Phantom 4A drone with one camera that can only detect a person [1]. The use of a CCTV camera is still an issue; the reason for this is that it is not in high resolution and has a limited viewing area [2]. To improve the previous research, unmanned aerial vehicles (UAVs) will be used, and two cameras will be employed: one for human detection and the other one is the thermal camera since they have the ability to identify the missing human as well as other people in the immediate vicinity and improve detection accuracy. When the drone detects a person, it will show up on

the FPV screen and identify the object with the use of YOLOv5. It can also capture any human that is recognized. People counters will be added wherein the number of persons detected for the live person count (LPC) and the overall person count (OPC) will be recorded, and the data gathered will reveal the total number of individuals detected on that day. The videos taken by the drones are automatically transferred and stored to electronic devices such as phones, tablets, and laptops.

The main objective of this study is to develop an unmanned aerial vehicle with human detection and people counter using YOLOv5 and thermal camera for search operations. Specifically, this study aims (1) to help in the search operations for persons who are in danger, (2) to create a prototype and program the UAV to automatically detects a person using YOLO v5 and count the live and total numbers of human detected, (3) to develop a GUI/software for the drone and the thermal camera, (4) to test the ability of the drone and the thermal camera to detect human.

This study will benefit rescuers as it helps them to react and respond much better in emergency situations such as natural disasters, fire rescue, and searching operations because the rescuers will know what the situation is and the drone can count the total people who are stranded in that certain area. The drone is programmed to detect humans, as well as count the live and overall detected people. The drone is also equipped with a thermal camera for accuracy. This study will help them in finding stranded people and rescuing will be much easier with this technology. After the disaster, when the weather gets better, the search operation will begin to determine if there are any persons trapped in a certain location.

II. METHODOLOGY

This section provides an overview of the capabilities of the unmanned aerial vehicle with human detection and people counter using YOLOv5 and Thermal Camera from the list of its components to the various processing techniques to keep track. The methodology gives an in-depth view of the conceptual framework, system processing flow, list of materials, prototype design, the graphical user interface, calibration of sensor, and statistical analysis of data.

A. Conceptual Framework

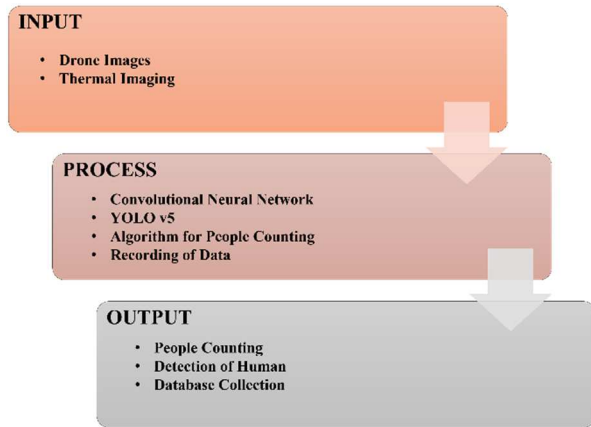


Fig. 1. Conceptual Framework

Figure 1 shows the conceptual framework of the study. The input for the human detection will be the two (2) cameras which are the thermal imaging and the image from the drone. For the process, it starts by data gathering which is the videos that the two (2) cameras will capture. While the data is being gathered, the system can already record the data using the GUI. The data gathered will undergo a convolutional neural network to identify if the object is human. The YOLOv5 algorithm was used to classify the object that was detected by the drone. If the system detects a person, it will undergo an application of a connected algorithm in the drone camera which is for people counting [3]. The output of the system includes detection of humans. Also, it counts all the humans that were detected by the system and the humans that were being detected in live. Lastly, all the captured images and videos of the thermal camera and drone camera will be saved to the laptop.

B. System Process Flow

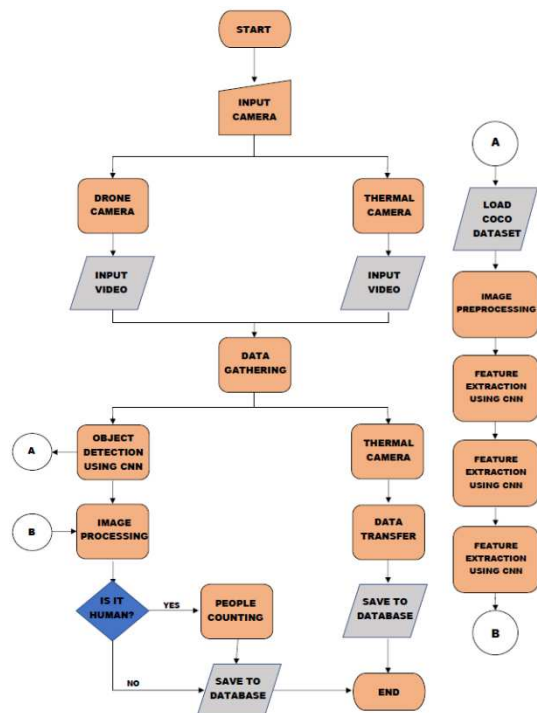


Fig. 2. System Process Flow Diagram

In Figure 2 the starting phase of the UAV detecting system is to attach the thermal camera to the lower part of the drone. The moment the camera displays the video capture using the GUI of each camera the data gathering will start immediately. The drone will be designed to recognize objects, while Deep CNN will be used to detect humans and the YOLOv5 algorithm for recognizing whether the object is human, and people counting will be added to the drone's features. For the secondary camera the Raspberry pi 3 will be programmed to operate the thermal camera and connect it to the second device using Wi-Fi for data transfer [4]. All the data collected will be saved in a database.

C. List of Materials

1. Tello Drone with built-in Camera
2. FLIR Lepton 2.5
3. Breakout Board
4. Raspberry Pi 3 Model B
5. Laptop
6. Battery
7. Battery Shield

D. Proposed Design and Setup



Fig. 3. Proposed Prototype



Fig. 4. Actual System Design

The sample prototype of the unmanned aerial vehicle with human detection and people counter using YOLO v5 and thermal camera is shown in Figure 3. It shows how each component of the device is connected to another. The thermal camera will be connected to the Raspberry Pi 3, which will then be positioned at the back of the drone alongside the battery shield. The rechargeable battery will be put in the battery shield to power the raspberry pi, and the raspberry pi and drone will be connected to the laptop through Wi-Fi to view the First Person View and be controlled. In figure 4, it shows the actual system design.

E. Graphical User Interface

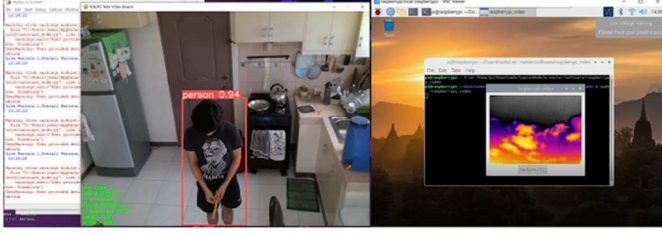


Fig. 5. Drone and Thermal Camera GUI

The Drone GUI is done by using Python, to control the camera and the aircraft functions. It displays a real-time view and shows if there is a human detected, the live and overall count, and the battery percentage. The next GUI is for the Thermal camera, or secondary camera, which is developed using the VNC Viewer. It also shows a real-time display as well as an accurate, calibrated, and noncontact thermal image output. This will help rescuers in determining whether the detected human from the drone camera is indeed a human. The GUI for Drone and Thermal camera are shown in Figure 5.

F. Calibration of Instruments or Materials

a. Training Process

For the training process, the dataset that was used by the researchers is the CoCo Dataset which is already a pre-trained model. It is a saved network that was previously trained on a large dataset, typically on a large-scale image classification task.

TABLE I. NUMBER OF HUMANS DETECTED IN DIFFERENT HEIGHTS

Distance Height	Number of Actual Counts of Human							
	1		2		3		4	
	Number of Detected Human	% Error	Number of Detected Human	% Error	Number of Detected Human	% Error	Number of Detected Human	% Error
500cm	1	0%	2	0%	3	0%	4	0%
1000cm	1	0%	2	0%	3	0%	4	0%
1500cm	1	0%	2	0%	3	0%	4	0%
2000cm	1	0%	2	0%	3	0%	4	0%
2500cm	1	0%	2	0%	2	33%	3	25%
Average Percentage Error: 2.9%								

Table 1 shows the data of object detection using YOLOv5 for detecting humans. This will be evaluated by determining the number of detected humans of the device in various distances comparing it to the actual number of humans. The percentage error will be calculated to determine the accuracy of the device. As seen in the results, the device was able to detect humans and obtained a low percentage error. The %Error will be calculated as shown in equation 2.1. The measured value will be the number of humans that were detected by the drone and the real value will be the number of actual counts of humans.

$$\text{Percent Error} = \frac{|\text{measured} - \text{real}|}{\text{real}} \times 100\% \quad (2.1)$$

III. RESULTS AND DISCUSSIONS

This section discusses the results and discussion of the test results of the human detection tests and the output of the drone and thermal camera at different distances and views.

TABLE II. DIFFERENT VIEWS OF ANGLE OF ONE PERSON

DIFFERENT VIEWS OF ANGLE OF ONE PERSON			DIFFERENT VIEWS OF ANGLE OF ONE PERSON		
Distance to Subject	Angle of View	Thermal Image Output	Distance to Subject	Angle of View	Drone Image Output
250cm	Front View		250cm	Front View	
	Back View			Back View	
	Top View			Top View	
	Side View			Side View	

Table 2 shows the output images of the prototype with different views. The device was tested in 250cm with different views such as front view, back view, top view, and side to check if it can detect the human for both cameras. As seen in the table for the output of the drone camera, it was able to detect it as a human and the device can detect a person even if only certain parts of the human are seen in the frame [5]. For the output of the thermal camera, it was also able to detect the human through its heat signature [6]. It can be seen in the camera that the human is detected as red, orange, and yellow colors.

TABLE III. COMPARISON OF WITH AND WITHOUT COLD MATERIAL

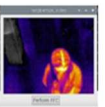
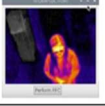
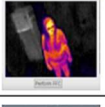
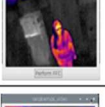
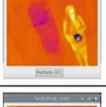
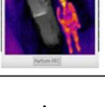
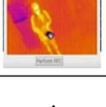
COMPARISON OF WITH AND WITHOUT COLD MATERIAL (ICE)			
Distance to Subject	Distance Height	Without Cold Material	With Cold Material
50cm	150cm		
100cm	150cm		
150cm	150cm		
200cm	150cm		
250cm	150cm		

Table 3 shows the comparison output images of with and without cold material for the thermal camera [7]. An ice served as the cold material in comparison to the output image without cold material. It is shown in the image that the FLIR Lepton 2.5 can detect the human within a distance of up to 250cm and with a height distance of 150cm. It can also observe that the cold objects detected by the thermal camera can be seen as white and humans can be detected as color yellow and orange.

TABLE IV. COMPARISON OF WITH AND WITHOUT HOT MATERIAL

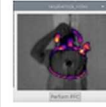
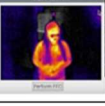
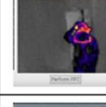
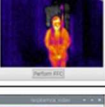
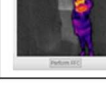
COMPARISON OF WITH AND WITHOUT HOT MATERIAL (KETTLE)			
Distance to Subject	Distance Height	Without Hot Material	With Hot Material
50cm	150cm		
100cm	150cm		
150cm	150cm		
200cm	150cm		
250cm	150cm		

Table 4 shows the comparison output images of with and without hot material for the thermal camera [8]. A kettle served as a hot material for this set up in comparison to the output image of without hot material. It shows that the human can still be seen with hot material within the range of 250cm, but it is not clear compared to without hot objects. It was observed that the color of the human that can be seen with the hot object is purple, and the hot material is orange which is the opposite in the output image of without hot material. It is because the color palette of the thermal camera automatically adjusts when it detects high temperature differences, and it depends on its environment.

TABLE V. IMAGES OF TESTING HUMAN DETECTION WITH SMOKE

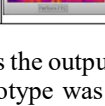
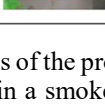
IMAGES OF TESTING HUMAN DETECTION WITH SMOKE				
Distance to Subject	Number of Actual Counts of Human			
	2		2	
	Thermal Image Output	Number of Detected Person	Drone Image Output	Number of Detected Person
50cm		2		2
100cm		2		2
150cm		2		2
200cm		2		2
250cm		2		2

Table 5 shows the output images of the prototype in a smoke setting. The prototype was tested in a smoke area. The smoke was generated by using a fog machine. Based on the data that was obtained, the thermal camera is able to detect humans in the smoke [9]. Two persons were able to detect by the thermal camera in the range of 150cm to 250cm. For the output of the camera of the drone, the two persons were able to detect by the prototype within the smoke in a given distance which can be used in a foggy area.

TABLE VI. IMAGES OF TESTING HUMAN DETECTION WHEN THE LIGHT IS TURNED OFF

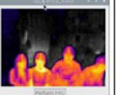
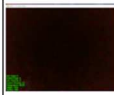
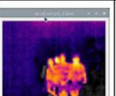
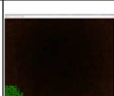
IMAGES OF TESTING HUMAN DETECTION WHEN THE LIGHT IS TURNED OFF				
Distance to Subject	Number of Actual Counts of Human			
	4		4	
	Thermal Image Output	Number of Detected Person	Drone Image Output	Number of Detected Person
100cm		4		0
100cm		4		0
300cm		4		0

Table 6 shows the output images of the prototype in a dark area [10]. This was tested in the house with the light switch off. As seen in the images, the camera of the drone was not able to detect any humans because it is not visible. For the thermal camera, it was able to detect all the humans at 100cm, 200cm, and 300cm. Lighting does not affect the performance of the camera since it only detects heat from the surroundings.

TABLE VII. IMAGES OF TESTING HUMAN DETECTION WITH MANNEQUIN

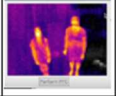
IMAGES OF TESTING HUMAN DETECTION WITH MANNEQUIN				
Distance to Subject	Number of Actual Counts of Human			
	2		2	
	Thermal Image Output	Number of Detected Person	Drone Image Output	Number of Detected Person
50cm		2		3
100cm		2		3
150cm		2		3
200cm		2		3
250cm		2		3

Table 7 shows the output images of the prototype with a mannequin. It shows in the output image for the drone that it detects three persons where there are only two actual humans. The device also detects the mannequin as an actual person. However, the output image for the FLIR Lepton 2.5 can only recognize two humans since the mannequin has a different temperature to actual humans.

IV. CONCLUSION

In conclusion, the researchers were able to design an unmanned aerial vehicle with human detection and people counter using YOLOv5 and thermal camera for search operations. This was performed by programming the Raspberry Pi 3 to display and control the thermal camera, which was then attached to the Drone as a second camera. The camera drone was programmed to detect humans and count people in real time using the YOLOv5 algorithm and the Coco Dataset, which was already pre-trained. The training method shows that the camera drone can detect humans at various distances and that the thermal camera can accurately determine humans at various distances using colors.

The first objective of this study is to help in search operations for persons who are in danger. This was accomplished by using unmanned aerial vehicles to help rescuers in responding much more quickly in emergency situations by allowing them to see what is going on in areas that are difficult to reach due to the dangerous nature of the area in order to determine if there are any people trapped.

The second objective is to create a prototype and program the UAV to automatically detect a person using YOLO v5 and count the live and total numbers of humans detected. This was achieved by testing the accuracy of both the drone camera and the thermal camera when detecting humans at various distances. The accuracy of the people counter is also evaluated; the percentage error is caused by the distance between humans being mistakenly identified as one.

The third objective is to develop a GUI/software for the drone and the thermal camera. This was done by creating a separate GUI for the drone and the thermal camera for display and controls. Python was used to display the drone's output image and operate it with keyboard keys for the controls. The captured images were saved to a specific directory, and the number of live humans and the total number of humans were entered into an excel file for database purposes. PuTTY and VNC viewer were used to program and display the thermal output image of the thermal camera, as well as to control for screen capture. Wi-Fi was used to connect the drone and thermal camera to the computers.

The last objective is to test the ability of the drone and the thermal camera to detect humans; this was achieved by testing the drone camera and thermal camera at different distances and situations. The drone can detect a mannequin or image as a human, but with the use of the thermal camera it can identify if the detected human is actually a human based on its thermal heat. Overall, the researchers were successful in achieving the study's specified and main objectives. Finally, the collected data proves the proposed design's reliability. All the data shows that the system can detect humans and can determine if they are

real humans using the thermal camera, as well as to count the live and overall detected humans.

V. RECOMMENDATION

The proposed system has many possible designs for other applications. To further improve the study, it is recommended to future researchers to use a smaller Raspberry Pi or larger drones to improve the drone's flight performance by minimizing the excessive weight imparted to the drone and by adding other functions such as animal detection and animal counting, the range of search operations can be expanded to include all living things. The researchers also recommend that future studies and testing be conducted in a higher location, such as mountains, in order to obtain more precise data on drone performance.

REFERENCES

- [1] S. Sambalek and M. Ivacic-Kos, "Automatic Person Detection in Search and Rescue Operations Using Deep CNN Detectors," in *IEEE Access*, vol. 9, pp. 37905-37922, 2021, doi: 10.1109/ACCESS.2021.3063681.
- [2] S. Powale, A. Dhanawade, S. Bagwe, S. Kawale, N. L. Chutke and S. Chavan, "Person identification in low resolution CCTV footage using deep learning," 2020 2nd International Conference on Advances in Computing, Communication Control and Networking (ICACCCN), 2020, pp. 236-240, doi: 10.1109/ICACCCN51052.2020.9362764.
- [3] S. R. C. De Guzman, L. C. Tan and J. F. Villaverde, "Social Distancing Violation Monitoring Using YOLO for Human Detection," 2021 IEEE 7th International Conference on Control Science and Systems Engineering (ICCSSE), 2021, pp. 216-222, doi: 10.1109/ICCSSE52761.2021.9545098.
- [4] A. C. Paglinawan et al., "Detection of three visual impairments: Strabismus, blind spots, and blurry vision in rural areas using Raspberry PI by implementing hirschberg, visual field, and visual acuity tests," 2017IEEE 9th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment and Management (HNICEM), 2017, pp. 1-9, doi: 10.1109/HNICEM.2017.8269550.
- [5] M. A. Y. Luberia, J. M. Quintos, J. C. Dela Cruz and E. D. Dimaunahan, "Preliminary Diagnosis for Flu using Facial Feature Recognition and Thermal Camera," 2021 5th International Conference on Electrical, Telecommunication and Computer Engineering (ELTICOM), 2021, pp. 128-133, doi: 10.1109/ELTICOM53303.2021.9590165.
- [6] Ahmad, M., Ahmed, I., Ullah, K., Khan, I., Khattak, A., & Adnan, A. (2019). Person detection from overhead view: A survey. *International Journal of Advanced Computer Science and Applications*, 10(4), 567–577. <https://doi.org/10.14569/ijacsa.2019.0100470>
- [7] "Seeing temperature through color - keeping warm," Beyond Penguins and Polar Bears. [Online]. Available: <https://beyondpenguins.ehe.osu.edu/issue/keeping-warm/>
- [8] "Your perfect palette," Teledyne FLIR. [Online]. Available: <https://www.flir.com/discover/ots/outdoor/your-perfect-palette/>. [Accessed: 09-Mar-2022].
- [9] "Can Thermal Imaging See Through Walls? And Other Common Questions | Teledyne FLIR." <https://www.flir.com/discover/cores-components/can-thermal-imaging-see-through-walls/> (accessed Jul. 20, 2021).
- [10] J. Barela, M. Kastek, K. Firmanty, and P. Trzaskawka, "Accuracy of measurements of basic parameters of observation thermal cameras," no. February 2018, 2016, doi:10.21611/qirt.2016.004.